Retire Confident
AUSTRALIAN RETIREMENT CALCULATOR

Understanding Your Results

A plain-English guide to reading and acting on your simulation output

Version 1.0

Contents

1. Why Your Results Are Ranges, Not Answers

The first thing to understand is what the calculator is actually doing. Rather than predicting the future, it runs hundreds or thousands of plausible futures — each with a different sequence of market returns — and asks: how often does your plan survive?

This is fundamentally different from a spreadsheet that calculates "your money runs out at age 83". That kind of single-path projection creates false precision. Real retirement involves good years, bad years, and (most dangerously) bad years early on that your plan may never recover from.

So when you see a success rate, a percentile band, or a chart that shows a fan of possible outcomes, that is not the calculator being vague — it is being honest about genuine uncertainty.

> 💡 *Think of the results as a weather forecast, not a train timetable. A 90% success rate doesn't guarantee success any more than a 90% chance of sunshine guarantees you won't get rained on — but it tells you whether to pack an umbrella.*

2. Monte Carlo: Success Rates and Percentile Bands

2.1 Display Modes — Nominal, Real, and Today $

The calculator offers three ways to display portfolio balances, spending, and income across the charts and Executive Summary. The underlying simulation always runs in nominal (future) dollars — the display mode is a presentation layer only and has no effect on the success rate, Age Pension calculations, or any other computed result.

Nominal $ shows values in the actual dollars of each future year — the number that would appear in your bank account or superannuation statement. A balance of $900,000 shown in year 25 means $900,000 of future purchasing power at that point in time, not today. This mode is useful when you need to compare the calculator’s output to a specific future dollar target, such as a Transfer Balance Cap limit or a projected account balance from your fund’s statement.

Real (Retirement Year $) removes the effect of inflation and expresses all values in the purchasing power of the year you retire. If you retire at 62, every year on the chart is shown in 62-year-old dollars. This is the most useful mode for assessing trends within the simulation — because inflation has been stripped out, a flat line genuinely means flat purchasing power, and a declining line means your standard of living is falling in real terms. Nominal charts can mask a quietly eroding plan behind rising dollar figures.

Today $ goes one step further, converting all values back to the purchasing power of the current calendar year. This is the most intuitive mode for people who are still working, because the numbers are directly comparable to your current salary, rent, and grocery bill. A projected spending figure of $58,000 in Today $ means roughly what $58,000 buys right now, regardless of when in retirement that year falls.

The difference between modes can be substantial. At 2.5% annual CPI, $1 of purchasing power today becomes approximately $2.09 in nominal terms after 30 years. A portfolio that looks like $1.2 million nominal at age 90 may represent only $575,000 in today's purchasing power. This gap is why the Real and Today $ modes often produce a more sobering picture than Nominal — and why they are generally more appropriate for long-horizon retirement planning.

When switching modes, all charts, the Executive Summary, and the Sensitivity Analysis table update simultaneously. The success rate displayed in the Executive Summary banner is unaffected — it reflects the proportion of simulations in which the portfolio remained solvent, which is a nominal calculation independent of how the results are presented.

2.2 What Is the Success Rate?

The success rate tells you: in what percentage of simulated futures did your money last the full retirement period?

A run of 1,000 simulations might produce 870 where your portfolio reached your target age with money remaining, and 130 where it ran out early. That gives a success rate of 87%.

Success Rate	Rating	What It Means
90%+	Excellent	Your plan is well-buffered. The main risk is dying with far more than you needed — worth checking whether you're enjoying enough.
85–90%	Very Good	Strong result with modest margin. Most planners would consider this acceptable. A bad sequence of returns (e.g., a crash in your first two years) is the main vulnerability.
75–85%	Moderate	Reasonable but not comfortable. One-in-four to one-in-six scenarios end in shortfall. Consider small adjustments — even $5–10k/year in reduced spending can shift this significantly.
<75%	Needs Work	More simulations fail than succeed in many unfavourable market regimes. This needs material changes: spending cuts, later retirement, part-time income, or some combination.

> 💡 *A 100% success rate is not the goal. It typically means you're being so conservative — either spending too little or assuming very low returns — that you'll almost certainly die with a large, unused surplus. Most financial planners aim for 85–90%.*

2.3 Longevity & Outcome Analysis Chart

When Monte Carlo or Historical Monte Carlo is active, a Longevity & Outcome Analysis chart appears below the Executive Summary. It translates the abstract success rate into a visual picture of where and why failures occur. Two tabs:

Longevity scatter: One dot per run. The X axis shows the simulation end age; the Y axis shows the final portfolio balance. Green dots are solvent runs; red dots are insolvent. Use this chart to diagnose the *character* of your plan's risk:

- A healthy plan shows a dense band of green dots across all ages, with the band rising to the right as longer-lived portfolios compound further. Any red dots are sparse and scattered randomly — failures are rare and not age-driven.
- A plan with **longevity risk** shows green dots at younger ages and red dots concentrated at older ages (say, beyond 88). The plan works well for an average lifespan but runs out of money if you live longer than expected. The fix is usually a modest spending reduction, a small increase in starting capital, or shifting a portion of assets to guaranteed income (annuity or pension).
- A plan with **sequencing risk** shows red dots spread across *all* ages — failures occur at 75, 82, and 95 alike. This means the plan is fragile to poor early returns regardless of lifespan. The fix requires structural changes: a larger sequencing buffer, a lower starting withdrawal rate, or a more defensive early-retirement asset allocation.
- A plan with **insufficient capital** shows most dots — green and red alike — clustered at younger ages, meaning the portfolio depletes quickly in most scenarios. This requires a fundamental reassessment of spending, retirement age, or starting balance.

The vertical spread of green dots at any given age shows the range of outcomes: a wide spread means return variance matters a lot, while a tight cluster means the plan's outcome is relatively insensitive to market conditions.

Ruin age distribution: A histogram of failed runs only, binned by the age at which the portfolio reached zero. This is more actionable than the overall success rate because it tells you *when* you are at risk, not just *whether* you are:

- **Failures concentrated after age 90:** The plan is structurally sound but exposed to tail longevity. A 3–5% spending reduction, or a deferred annuity starting at 85, would likely eliminate most failures. The cost of fixing this is low.
- **Failures concentrated between ages 80–90:** Moderate structural shortfall. Consider a combination of modestly lower spending, a slightly later retirement, or a higher allocation to growth assets in early retirement to improve compounding. Review whether the Age Pension offsets are being fully captured.
- **Failures concentrated before age 80:** The plan has a fundamental capital or spending problem that cannot be solved by longevity hedging alone. The starting withdrawal rate is likely too high. Prioritise reducing spending, increasing the starting balance, or delaying retirement — or some combination of all three.
- **An empty histogram (no bars):** All runs succeeded. This is a good outcome, though if the success rate is 100% it may indicate the plan is overly conservative and you are likely to leave a large unspent surplus.

When stochastic mortality is enabled, the X axis of the scatter will show genuine spread reflecting the distribution of sampled lifespans. When it is off, all dots share the same end age (the fixed Modelled Death Age), so the scatter becomes a vertical column — informative about the range of final balances at that age, but not about longevity risk specifically.

2.4 Understanding P10, P50, and P90

When Monte Carlo mode shows percentile bands on the portfolio balance chart, it is showing you three representative outcomes drawn from your full set of simulations:

Band	Meaning	How to Use It
P10	Worst 10% of outcomes	The uncomfortable question: can you live with this? If the P10 line runs out of money at 78, that's a one-in-ten chance of serious financial difficulty. This is your key planning number — not P50.
P50	Median — the middle outcome	Half of simulations did better, half did worse. This represents your most likely trajectory, but 'most likely' still means a coin-flip between better and worse.
P90	Best 10% of outcomes	A lucky scenario — strong early returns, no major shocks. Do not build your retirement around this line. It's shown so you can see the upside, not plan for it.

The gap between P10 and P90 is your range of uncertainty. A very wide fan indicates high sensitivity to sequence-of-returns risk — meaning an unlucky run of bad years early in retirement has an outsized impact compared to the same bad years occurring later.

A note on income and spending bands when Variable death age is enabled

When stochastic mortality is active, each parametric Monte Carlo run ends at a sampled death age rather than a fixed horizon. This affects how the income and spending percentile bands are computed. At each year, only runs that are still active contribute to the income and spending bands — runs that have already ended because the simulated person died are excluded rather than counted as zero. This is intentional: a run that ended at age 83 should not pull the median income to zero for runs where the person is still alive at 84.

The practical effect is that the income and spending bands narrow progressively at older ages as fewer runs remain active, and they terminate entirely when the last active run ends. The balance bands behave differently — completed runs contribute their final balance to the calculation rather than being excluded, so the P50 balance line remains meaningful throughout the full chart period.

If the income and spending bands appear to end earlier than you expect, this reflects the distribution of sampled death ages rather than portfolio failure. The Longevity scatter tab in the Longevity & Outcome Analysis chart shows the spread of end ages across all runs, which gives you a direct picture of where the band data thins out.

2.5 Sequence-of-Returns Risk: Why Timing Matters

One of the most important concepts in retirement planning is that a 10% market drop at age 65 is far more damaging than the same 10% drop at age 75. Here's why:

- **At 65,** you are drawing down the portfolio to fund spending. A crash forces you to sell more units at depressed prices to raise the same cash. Those sold units can't recover when markets rebound.
- **At 75,** your portfolio is smaller anyway, and you have fewer years for a bad run to compound into a catastrophe.

This is why a bad sequence of returns in the first few years of retirement is far more damaging than the same bad returns occurring later — and why the sequencing buffer (cash reserve) and formal stress tests B1 (Crash) and B2 (Bear Market) are particularly important to run. Note that the fan of outcomes on the chart will typically widen over time as uncertainty compounds; this is normal and expected, not a cause for alarm.

> 💡 *If your P10 line drops sharply in the first 5–8 years of retirement, that is a sequence-of-returns problem. Solutions include: a sequencing buffer, dynamic guardrails, or delaying retirement by 1–2 years.*

2.6 Historical Monte Carlo vs Parametric Monte Carlo

The calculator offers two Monte Carlo modes, and they answer slightly different questions:

- **Parametric Monte Carlo** generates random returns using your specified expected return and volatility. It explores a wide mathematical range, including scenarios that have never actually occurred. Use it to understand sensitivity to your assumptions.
- **Historical Monte Carlo** samples from 98 years of real market data (1928–2025). These are returns that actually happened — including the Great Depression, World War II, 1970s stagflation, the Dot-com crash, and the GFC. It cannot simulate things outside that range, but every scenario it produces has real-world grounding.

For serious retirement planning, Historical Monte Carlo (Block Bootstrap method, block size 5–7 years) is the most credible analysis available in the calculator. It naturally captures the multi-year correlations that make crashes painful — bad years tend to cluster.

3. Reading the Annual Spending Breakdown Chart

The Annual Spending Breakdown chart is arguably the most diagnostic chart in the calculator. Rather than just showing you whether you survived, it shows you how — and reveals structural vulnerabilities that the summary metrics can miss.

3.1 What the Colours Mean

Colour	Source	What to Look For
Blue	Main super	Your primary drawdown — the dominant bar in early retirement.
Yellow	Cash account	Appears if you have a cash component. Usually small.

Colour	Source	What to Look For
Orange	Sequencing buffer	Draws down early in retirement if you have a buffer configured.
Dark green	Age Pension	Centrelink payments. Watch for this growing as super depletes.
Medium green	Defined Benefit / Annuity	Stable income floor. A large green segment here is a powerful safety net.
Light green	Other income	Part-time work, rental, etc. — usually present in early retirement only.

3.2 Healthy vs. Concerning Patterns

Below are the key patterns to recognise:

Pattern 1: Gradual Transition — Healthy

In early retirement, the bars are dominated by blue (super). Over time, as super depletes, the dark green (Age Pension) grows to fill the gap. By late retirement, Age Pension and defined benefit income cover most spending. This is the expected, healthy pattern — the system is working as designed.

Pattern 2: Blue Disappears Abruptly — Warning

If the blue super bar disappears suddenly (rather than gradually) in mid-retirement, it means super ran out in those years. Check what fills the gap: if Age Pension and other income are sufficient, the plan may still be viable. If there's a shortfall (total bar height drops below your spending line), you have a real problem.

Pattern 3: Heavy Age Pension Reliance from Day One — Investigate

If Age Pension accounts for a large portion of income from the very start of retirement, your super balance may be lower than ideal relative to your spending, or you're entering retirement old enough that Age Pension kicks in promptly. This is not automatically bad, but it means your lifestyle is more exposed to any policy changes to the Age Pension.

Pattern 4: Spending Line Drops in Later Years — Intentional or Not?

The grey dashed line shows the minimum required superannuation drawdown. If your total spending (the bars) drops noticeably below your intended spending in later retirement, check whether the JP Morgan declining spend curve is active — this models the research finding that real spending naturally falls in late retirement as mobility decreases. If that's the cause, it's a deliberate modelling choice. If it's unexpected, investigate.

> 💡 *The Annual Spending Breakdown is best read in 'Real $' mode (toggle at the top of the screen). This removes the illusion of growing bars caused by inflation and shows you actual purchasing power each year.*

3.3 What a Solid Income Floor Looks Like

The most resilient retirement plans have a meaningful floor of non-super income — defined benefit pension, Age Pension, or annuity income — that covers a substantial portion of basic living costs. When that floor exists, market downturns have less power to derail the plan because essential spending doesn't depend on investment performance.

If your income floor (green bars) covers 50–70% or more of your spending in late retirement, your plan is structurally robust. If it covers less than 30%, your lifestyle in late retirement is heavily dependent on investment returns over the next 20–30 years — worth stress-testing carefully.

4. Reading the Formal Stress Test Results

The Stress Tests mode runs your plan through eight predefined scenarios, each designed to expose a specific type of vulnerability. No real retirement will exactly match any of these scenarios — but a robust plan should be able to absorb most of them.

Rather than looking at each test in isolation, read them as a group. The pattern across tests tells you far more than any single result.

4.1 What Each Test Is Probing

Test	Name	Scenario	What Failure Reveals
A1	Base Case	5% return, 2.5% inflation — a moderate, steady environment	If A1 fails, your base spending is simply too high for your super balance. Start here.
A2	Low Returns	3.5% return — a world of structurally lower investment returns	Sensitivity to a prolonged low-growth environment (e.g., Japan post-1990). More likely than people assume.
B1	Crash	Immediate -25% crash, then 5% recovery	Sequence-of-returns risk in the first year. Most dangerous event is a major crash just after retirement.
B2	Bear Market	10 years of zero real returns, then 5%	Sustained mediocrity — the slow grind that slowly depletes super without the drama of a crash.
B3	High Volatility	Wild swings averaging 5%	Whether the average return matters less than its distribution. Tests guardrail effectiveness.
C1	High Inflation	5% inflation across the full period	Purchasing power erosion. Particularly damages fixed-income assets and spending that can't scale down.
D1	Extreme Longevity	Survival to age 105 — a 45-year retirement	Whether you're genuinely planning for longevity or implicitly assuming average life expectancy.
G1	Health Shock	$30k/year extra medical costs from age 75	Late-retirement healthcare costs that many people underestimate, particularly relevant for singles.

Test	Name	Scenario	What Failure Reveals
H1	Worst Case	Crash + high inflation + health costs combined	The 'perfect storm'. A plan that survives H1 is genuinely robust. Most plans don't need to.

4.2 How to Read Your Stress Test Results Together

Work through the following questions after running the formal tests:

- **Does A1 succeed?** If not, stop here — this is the most urgent problem. Your base spending exceeds what your super can sustain even with steady, moderate returns. Reduce spending or delay retirement before proceeding.

- **Does A2 succeed?** If A1 succeeds but A2 fails, your plan depends on investment returns remaining at or above the current Australian market average. That's not unreasonable, but be aware of the risk.

- **Do B1 and B2 differ significantly?** If B1 (crash) fails but B2 (bear market) succeeds — or vice versa — that tells you whether your vulnerability is to sharp shocks or grinding underperformance. A sequencing buffer helps with B1; guardrails help more with B2.

- **Does C1 (high inflation) cause notable deterioration?** If yes, your spending is not adequately indexed and your non-CPI income is significant. Defined benefit pensions with fixed (non-CPI) indexation are particularly exposed here.

- **Does D1 (longevity) fail materially earlier than your planned endpoint?** This is worth examining carefully. A plan that runs to 90 but fails at 97 is not a problem if you're confident your family longevity doesn't extend that far — but worth discussing with a financial planner if you have long-lived parents.

- **Does G1 (health shock) fail?** If yes, consider whether you have other health-cost buffers — private health insurance, savings earmarked for care, or the ability to scale back spending. Aged care modelling (built into the calculator) is worth enabling separately.

> 💡 *A robust plan should survive A1, A2, B1, B2, and C1 under most circumstances. B3, D1, and G1 are secondary — their failure indicates specific vulnerabilities worth noting but not necessarily fixing at the cost of current quality of life. H1 is a worst-case scenario: surviving it is a bonus, not a requirement.*

5. Age Pension: When It Kicks In and What It Means

The Age Pension is Australia's most under-appreciated retirement asset. Many people approaching retirement underestimate how significant it will become — particularly in late retirement when super balances have declined.

5.1 How Eligibility Works

The Age Pension becomes available at age 67 (current qualifying age for anyone born after 1 January 1957). Eligibility is then determined by two means tests, both applied every fortnight — and whichever test produces the lower payment is the one that applies:

- **Assets Test:** Centrelink assesses your total assets (excluding your principal home if you own one). For every $1,000 of assets above the threshold, your pension reduces by $3/fortnight. As your super draws down over retirement, your assets fall — and your pension typically increases.

- **Income Test:** Centrelink assesses your income from all sources including deemed income from financial assets (even if you don't actually earn that amount). Deeming rates set a notional income for assets regardless of actual returns.

The calculator models both tests automatically, using the inbuilt thresholds and rates.

5.2 The Natural Increase Pattern

One of the most important — and often counterintuitive — dynamics in the chart is that Age Pension payments typically grow over time, even before accounting for indexation. This happens because your super balance is declining:

- **Years 1–10:** Super is high, so you're likely above the assets test threshold for a full pension. You may receive a part pension or nothing at all if assets are very high.

- **Years 10–20:** As super depletes below the threshold, your entitlement grows — sometimes significantly. Many retirees move from a small part-pension to a substantial payment in this phase.

- **Years 20+:** With super largely or fully depleted, Age Pension may become your primary income source alongside any defined benefit pension.

This pattern means the Age Pension acts as a natural floor that rises as your investment portfolio falls — a form of built-in insurance against longevity that many people don't fully appreciate when they're planning at age 55 or 60.

5.3 Couples vs Singles

Age Pension rates for couples are not simply double the single rate — they are approximately 1.5x the single rate. This means:

- **When modelling as a couple,** your combined maximum Age Pension is roughly 75% of two single pensions, not 100%. The assets test thresholds are higher for couples, but not proportionally so.

- **Death scenarios matter here.** When a partner dies, the survivor reverts to single-rate Age Pension. This can be a significant income reduction at a time of heightened expenses and emotional difficulty. Use the death scenario modelling to check the surviving partner's position.

5.4 What the Work Bonus Does

If you or your partner continue to earn employment or self-employment income in retirement (part-time work, consulting), the Work Bonus allows up to $11,800 per year (indexed) of employment income to be excluded from the income test. This can meaningfully increase Age Pension entitlement for those who do some paid work in early retirement.

If you plan to do some part-time work in early retirement, enter it under Other Net Income and tick the "Employment income — Work Bonus eligible" checkbox. This tells the calculator to apply the Work Bonus concession, which partially shields employment income from the Age Pension income test. The interaction is often more favourable than people expect — enter the income and observe the effect on your Age Pension entitlement in the results.

> 💡 *If you're planning to do some part-time work for the first 5 years of retirement, enter it in the Other Net Income section and observe the impact on Age Pension. The interaction is often more favourable than people expect, because the Work Bonus partially shields employment income from the income test.*

6. The Ending Balance: How Much Is Enough?

The ending balance — what remains when the simulation reaches your target age — is one of the most misunderstood metrics in retirement planning. More is not always better. The goal is not to maximise the ending balance; it is to maintain your desired lifestyle for your full retirement with an acceptable margin of safety.

6.1 What a Large Ending Balance Actually Means

If your P50 (median) simulation ends with a very large positive balance — say, 150% or more of your starting super — that suggests one or more of the following:

- **You're spending too little.** Your quality of life in retirement is lower than it needs to be. You could afford to increase spending, retire earlier, take more holidays, or give more to family now while you can enjoy seeing the benefit.
- **Your return assumptions are optimistic.** If you've set an 8% return assumption with 15% volatility, the median outcome will look very comfortable — but the downside scenarios (P10) may be much less rosy. Check whether P10 also shows a large positive balance before concluding you're over-saving.
- **You have a legacy goal.** If you intend to leave an estate, a large ending balance is intentional. The calculator doesn't make this explicit — it's worth entering it as a target ending balance in your mental model.

6.2 Interpreting Zero or Near-Zero Endings

A plan that ends at exactly $0 is, mathematically, a perfect use of resources. In practice:

- **In the P50 scenario,** a near-zero ending is acceptable — it means the median outcome is efficient. But check P10: a P10 line that hits zero at age 78 is a serious problem even if P50 looks fine.

- **In the P10 scenario,** running out before age 85 is a genuine risk. Running out after 90 is more acceptable, as the probability of reaching that age and still needing to fund a full lifestyle is relatively low.

6.3 A Practical Framework for Evaluating Your Ending Balance

P50 Ending Balance	P10 Ending Balance	Interpretation
Large positive (>80% of start)	Positive	Consider whether you're under-spending. You may be sacrificing quality of life unnecessarily.
Moderate positive (20–80% of start)	Positive or near zero	Well-calibrated. Good margin without excessive over-saving.
Small positive or near zero	Near zero or runs out after 85	Efficient but tight. Guardrails recommended. Keep reviewing annually.
Runs out before target age	Runs out significantly early	Under-resourced. Needs adjustment to spending, retirement age, or income.

6.4 The Legacy Question

If leaving money to children, grandchildren, or charity is a goal, the ending balance matters in a different way. The calculator doesn't have an explicit bequest input, but you can model it by:

- **Setting a higher target age** (e.g., 100 rather than 90) to ensure money lasts well beyond expected life expectancy, leaving a de facto estate.

- **Checking the P10 ending balance** — if P10 is still positive at your target age, you have a high probability of leaving something regardless of market conditions.

- **Noting that Age Pension assets test thresholds** mean you can often hold more in the family home (which is exempt) and deplete super more aggressively, while maintaining Age Pension eligibility. This can actually increase wealth transfer compared to holding excess super.

> 💡 *The principal home is exempt from the Age Pension assets test. If you own a valuable property and have moderate super, you may qualify for a higher Age Pension than you expect — and your estate can still include the property value.*

6.5 Finding Your Sustainable Spending Level

6.5 Finding Your Sustainable Spending Level

The sustainable spending figure is the most direct answer the calculator can give to the central retirement planning question: how much can I actually afford to spend?

The Constant Return result gives you a deterministic ceiling — the maximum spending where your portfolio reaches exactly your target ending balance at the Modelled Death Age, assuming returns are steady every year. Think of it as the upper bound under ideal conditions. If your current spending is already above this number, the plan has a structural problem regardless of market conditions. If it is well below, you may be sacrificing lifestyle unnecessarily.

The Monte Carlo result is more meaningful for real planning. It tells you the maximum spending where a chosen percentage of simulated futures still end successfully — for example, "at $95,000/year, 80% of 1,000 market simulations left me solvent." This number is lower than the constant-return result because it builds in a buffer against bad luck: a market crash in your first two years, a prolonged period of low returns, or an unlucky sequence of expenses. The gap between the two figures is itself informative — a wide gap indicates your plan is highly sensitive to sequence-of-returns risk and the constant-return figure is a poor guide to actual safe spending.

The sensitivity strip alongside the MC result shows the success rate at $5,000 above and below the answer. A steep drop — say, 80% at $95k, 72% at $100k — tells you that modest overspending materially increases your risk. A shallow slope — 80% at $95k, 78% at $100k — tells you there is genuine room to spend a little more without much consequence.

When the result surprises you in either direction, treat it as a signal: a much lower figure than you expected suggests your current plan is fragile and needs attention; a much higher figure suggests you are being unnecessarily conservative and could improve your quality of life.

7. Common Result Patterns and What They Diagnose

Experience with this type of modelling reveals that certain result patterns appear repeatedly. Recognising them helps you move quickly from 'what do I see?' to 'what should I do?'

Pattern A: "Cliff Edge" — Good Until It Isn't

The portfolio holds steady for 15–20 years, then drops sharply to zero. This is the most common failure pattern and has a specific cause: you're drawing down super at a rate that is sustainable in the short term but erodes capital faster than returns replenish it. The "cliff" is the point where the compound depletion becomes too fast to arrest.

Diagnosis: Spending is slightly too high relative to super, OR Age Pension is not being modelled (or is delayed beyond when it would realistically kick in). Check that "Include Age Pension" is enabled and that assessable assets are correctly entered.

Pattern B: "Good Middle, Rough Ends" — The Monte Carlo Fan

The P50 line is fine, the P90 line is very comfortable, but the P10 line runs out 10–12 years early. This is a healthy plan with sequence-of-returns vulnerability.

Diagnosis: The plan is sound in average and good conditions, but a run of bad luck early in retirement could be damaging. Solutions: add a sequencing buffer ($50–150k in a defensive cash/bond account), enable guardrails, or hold 1–2 years of spending in cash at retirement.

Pattern C: "Survivor" — Large Surplus Despite Stress Tests

All formal tests succeed, the Monte Carlo P10 is positive, and the ending balance is very large. This is genuinely a comfortable position — but check whether you're leaving quality of life on the table.

Diagnosis: Either your super is generous relative to your spending, or you have a strong income floor (defined benefit, rental income). Ask the question: if I increased annual spending by $15–20k, does the plan still pass the tests that matter (A1, A2, B1, B2)? If yes, consider spending more.

Pattern D: "Works on Paper, Fails Under Stress" — The Brittle Plan

A1 (base case) succeeds comfortably, but B1 (crash) and B2 (bear market) both fail, and the Monte Carlo success rate is below 80%.

Diagnosis: The plan only works if returns are steady. This is the most dangerous profile because it feels fine in optimistic scenarios but has material failure risk in realistic ones. Solutions: reduce spending, add a sequencing buffer, or delay retirement by 1–2 years to build more super.

8. Sensitivity Analysis: Testing Return and Spending Assumptions

The Sensitivity Analysis panel appears automatically in the results column when you select Constant Return — no separate button is required. Click the “Sensitivity Analysis — Return vs Spending” header to expand it.

It answers a different question from Monte Carlo: not “what is the range of probable outcomes given market volatility?” but “how sensitive is my plan to my assumptions about investment returns and spending?” These are two distinct risks, and both matter.

The Sensitivity Analysis panel is only visible when Constant Return is the active scenario. It is hidden in Monte Carlo, Historical Monte Carlo, Stress Test, and Historical Period modes — those modes model uncertainty directly, making a deterministic return grid redundant. If you cannot see the panel, switch to Constant Return mode.

8.1 What the Grid Shows

The table is a 7×5 grid. Rows represent investment return rates — your current return setting plus and minus up to three percentage points in one-point steps. Columns represent spending levels at 80%, 90%, 100%, 110%, and 120% of your base spending, with the actual dollar amount shown beneath each percentage. Your current settings are highlighted with a blue border so you can instantly locate where you sit in the grid.

Each cell shows the portfolio balance at the modelled death age. If the portfolio survived, it shows the ending balance. If it ran out, it shows “Depleted yr X” — the year the money was exhausted. All your other settings (Age Pension, income streams, defined benefit pension, aged care, couple tracking, debt, splurge spending, and inflation) are included in every cell. Only return rate and base spending vary across the grid.

8.2 Reading the Colour Coding

Colour	What It Means
Dark green	Survived to the modelled death age with more than $1M remaining. Comfortable outcome under this combination of return and spending.
Light green	Survived to the modelled death age with $500k–$1M remaining. Solid outcome — plan held up under this scenario.
Yellow	Survived to the modelled death age but less than $500k remaining. Plan worked, but with limited margin.
Orange	Portfolio depleted, but lasted beyond year 25. A late-retirement shortfall — less severe than early depletion, but still a concern for longevity planning.
Red	Portfolio depleted before year 25. A significant early shortfall requiring material plan changes.

> 💡 *Look at how far the green zone extends from your blue cell. If green fills most of the rows above and below your current return, your plan is robust to return assumptions. If you're sitting at the boundary of orange or red, your plan is sensitive to getting your return assumption right — worth running Monte Carlo to understand the full distribution of risk.*

8.3 Why the Blue Cell May Differ from the Executive Summary

If you have run a Monte Carlo or Historical simulation, you may notice the result in the blue cell does not match the figures in the Executive Summary. This is expected and not an error. The Executive Summary reflects the median of many simulations incorporating market volatility. The blue cell always shows a single deterministic run at your current return rate — no volatility, no randomness — which is the closest equivalent to the Constant Return view. The two tools are answering different questions and should be read alongside each other, not compared as if they should agree.

8.4 What the Sensitivity Analysis Does Not Show

Two important limitations to be aware of:

- **No market volatility.** Each cell uses a constant return every year. A 7% row does not mean 7% average with good and bad years around it — it means exactly 7% every single year. This eliminates sequence-of-returns risk entirely. A plan that looks green in the sensitivity grid can still fail in Monte Carlo due to a bad run of early returns. Use both tools together.

- **No stochastic irregular expenses.** If you have stochastic irregular expenses enabled, the sensitivity grid does not vary them across cells. Each cell uses a fixed deterministic path for those costs. The grid is most accurate as a tool for understanding return and spending sensitivity, not for modelling the full range of irregular expense outcomes.

9. Quick Reference: What Am I Looking At?

I See...	What It Means / What to Do
Success rate < 75%	Material risk. Adjust spending, retirement age, or income before finalising your plan.

I See...	What It Means / What to Do
Success rate 75–85%	Moderate. Small adjustments can move this to the green zone. Try reducing spending by 5–10%.
Success rate > 90%	Comfortable. Check whether P10 is also positive — if yes, consider whether you could spend more.
P10 runs out before 85	Sequence-of-returns risk is real. Add a buffer, enable guardrails, or reduce spending.
P10 survives to target age	The uncomfortable 10% of scenarios still end okay. Your plan has real robustness.
A1 fails	Base spending too high. This is the highest-priority problem to fix.
A1 passes but B1/B2 fail	Sequence-of-returns and/or sustained low-return vulnerability. Add buffer or guardrails.
Large blue bars throughout	Heavily reliant on super drawdown. Check what happens if super runs out earlier than expected.
Age Pension grows strongly mid-retirement	Working as designed — the natural income floor rising as super depletes. Healthy pattern.
Very large ending balance	You may be under-spending. Run a scenario with 10–15% higher spending and check the results.
Wide fan (P90 vs P10 very far apart)	High sensitivity to return sequence. Focus on P10, not P50. Consider defensive strategies.